Master Multimedia Networking: Content Distribution Networks, MN905

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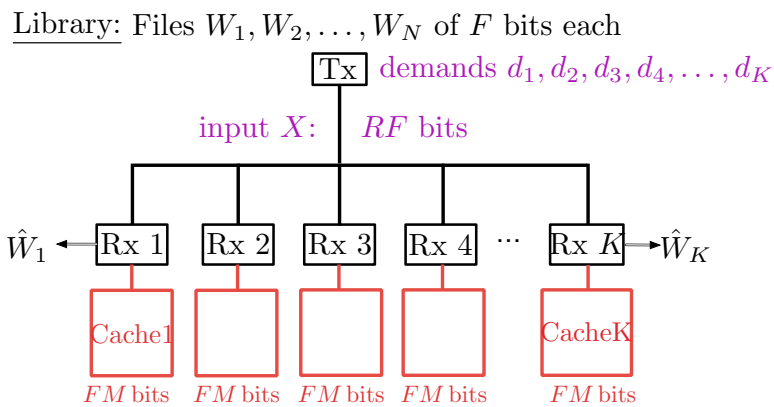
Written Exam, 15 December 2021

Instructions: You have 60 min to solve this part of the exam. All written documents (including books etc.) are allowed. No electronic devices.
We encourage you to look at all the questions. You might find some of them easier than others. Moreover, sometimes you might be able to answer for example question 3.) even if you could not answer the previous question 2.). Notice also that at any given time in the exam you can use all the statements that you were asked to prove in previous parts of the exam, even if you were not able to prove them.
Make sure you justify your answers! Your answers are graded also according to your justifications.

Problem Description:

Consider the following network with a single server and K receivers, where each receiver has a cache memory of size MF bits. Delivery communication X, consisting of RF bits, takes place one every receiver has emitted its demand. Storage in cache memories has to be performed without knowing the demands $d_1, \dots, d_K$. Each file in the library consists of F independent and identically distributed bits. The goal is that at the end of the delivery communication each receiver k can reconstruct its demanded file W_{d_k}.



Answer the following questions:

1. Explain the difference between centralized and decentralized caching. *(1pts)*

2. Is the fundametal rate-memory tradeoff $R^\star(M)$ smaller for centralized or decentralized caching systems? Justify your answer! *(1pts)*

3. Is the fundametal rate-memory tradeoff $R^\star(M)$ non-decreasing or non-increasing in the number of users K in the system? Justify your answer! *(1pts)*

4. Eve convinces you that she has a scheme that on a system with $K = 4$ users and $N = 10$ files requires cache size $M = 5$ and delivery rate $R = 0.55$. What can you deduce about the fundametal rate-memory tradeoff $R^\star(M = 5)$ of this system for $M = 5$? Is it larger or smaller than 0.6? *(1pts)*

Starting from here, assume a centralized caching system.

5. Consider the standard coded caching system with $K = 4$ users and $N = 40$ files. Describe a scheme that achieves rate-memory pair $(R = 2.75, M = 5)$. *(1pts)*

6. Assume that $K = 10$ and $N = 10$. For each of the following rate-memory pairs say whether they are achievable or not in the standard coded caching system: *(5pts)*

   (a) $M = 0.5$ and $R = 4$

   (b) $M = 5.3$ and $R = 0.65$.

   (c) $M = 9$ and $R = 0.1$.

   (d) $M = 9$ and $R = 0.99999$.

7. Consider now the standard coded caching system in above figure with $N = 20$ and $K = 2$, but where only receiver 1 has a cache memory but not receiver 2. Also, assume that the perfect link used for delivery communication is replaced by an erasure broadcast channel where receivers 1 has erasure probability 0.2 and receiver 2 has erasure probability 0.1. Assume that $M = 5$. What is the smallest R that you can find such that $(M = 5, R)$ is an achievable rate-memory pair. In other words, present the best possible placement and delivery scheme that you can think of so that R is minimized. (The smaller R, the more points you will get for your answer.) *(3pts)*